

PROFESSIONAL DATA ANALYSIS REPORT

Customer Demographic & Financial Analysis

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Project Exploratory Data Analysis (EDA)

PROJECT OVERVIEW



Objective

Analyze customer demographic and financial data to uncover critical patterns for banking and credit risk assessment.



Dataset

50,000 Records featuring variables: Age, Annual Income, Credit Score, and Occupation Status.



Techniques

Exploratory Data Analysis (EDA) using distribution modeling, correlation mapping, and segment profiling.

Demographic Analysis

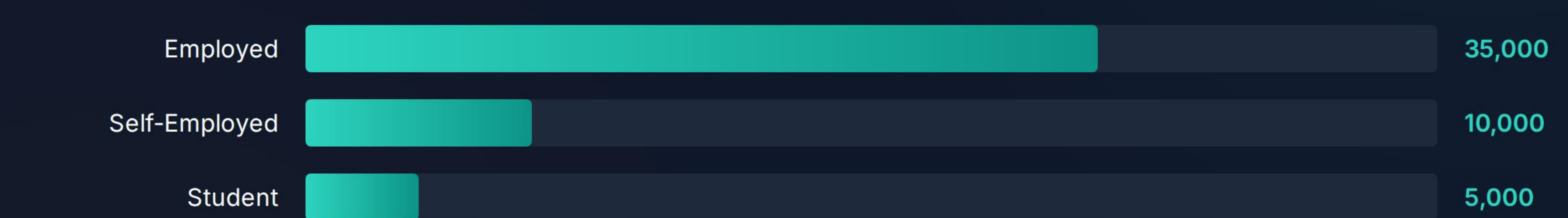
Mapping customer age and employment status segments

AGE DISTRIBUTION: MARKET CONCENTRATION



- ✓ **Prime Segment:** Majority of customers are aged between 25 and 45 years.
- ✓ **Growth Opportunity:** Young adults form the largest individual customer cluster.
- ✓ **Insight:** Minimal representation of users over 60, suggesting a digital-first or professional-heavy base.

| OCCUPATION STATUS PROFILE

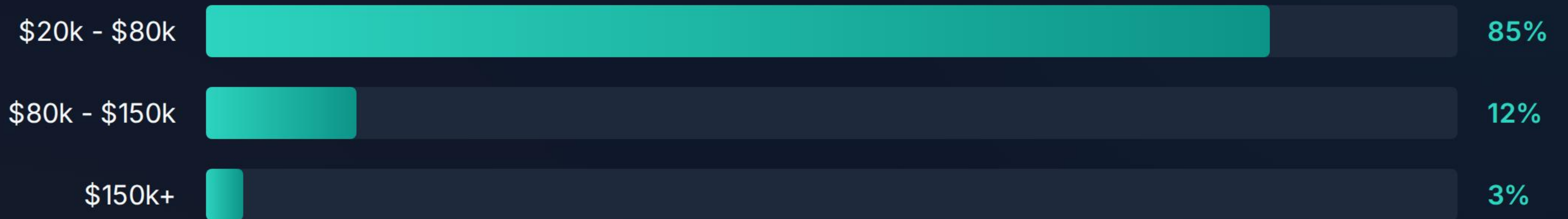


Strategic Note: Loan eligibility and credit limits should be heavily indexed against the dominant 70% employed segment.

Financial Analysis

Income variance and creditworthiness distribution

ANNUAL INCOME: SKEWED WEALTH



Right-Skewed Distribution: Most earners sit in the \$20,000 - \$80,000 range.

Premium Target: A small "long tail" of high-income individuals presents a premium product opportunity.

CREDIT SCORE: DISTRIBUTION BASELINE

- ✓ **Normal Distribution:** The credit score profile follows a healthy bell curve.
- ✓ **Mean Concentration:** Most customers fall within the 600 - 700 range.
- ✓ **Risk Assessment:** Rare occurrences of extremely low or high scores indicate a balanced, mid-risk portfolio.



CORRELATION: INCOME VS. CREDIT SCORE

 Income vs Credit Scatter Plot



Key Finding: Weak Positive Correlation

- ✓ Higher income alone does not guarantee a high credit score.
- ✓ Creditworthiness is likely driven by payment history, debt ratio, and age of accounts rather than gross salary.
- ✓ **Action:** Don't use income as a single proxy for risk.

KEY STRATEGIC INSIGHTS



Dominant Segment

Working-age (25-45) professionals dominate the portfolio. Marketing should focus on career-growth financial products.



Risk Profiling

A balanced 600-700 credit score concentration suggests a stable lending environment with moderate risk tolerance.



Data-Driven Upsell

Identify high-income outliers for premium wealth management services and high-limit credit offerings.

STRATEGIC CONCLUSION

- ✓ **Segmentation:** Occupation status is a powerful lever for loan eligibility tailoring.
- ✓ **Predictive Analytics:** The weak income/credit link necessitates multi-factor scoring models.
- ✓ **Stability:** Findings indicate a stable customer base with high growth potential in younger tiers.



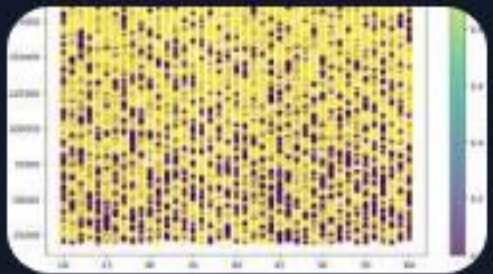
THANK YOU

Any Questions?

Strategic Financial Analysis & Demographics Report

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IMAGE SOURCES



<https://cdn.ncbi.nlm.nih.gov/pmc/blobs/5367/12876229/c279f9f6db3c/frai-08-1718854-g005.jpg>

Source: pmc.ncbi.nlm.nih.gov



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